

50. A trucker handed in a ticket at a toll booth showing that in 2 hours she had covered 159 mi on a toll road with speed limit 65 mph. The trucker was cited for speeding. Why?
51. Classical accounts tell us that a 170-oar trireme (ancient Greek or Roman warship) once covered 184 sea miles in 24 hours. Explain why at some point during this feat the trireme's speed exceeded 7.5 knots (sea miles per hour).
52. A marathoner ran the 26.2-mi New York City Marathon in 2.2 hours. Show that at least twice the marathoner was running at exactly 11 mph, assuming the initial and final speeds are zero.
53. Show that at some instant during a 2-hour automobile trip the car's speedometer reading will equal the average speed for the trip.
54. **Free fall on the moon** On our moon, the acceleration of gravity is 1.6 m/sec^2 . If a rock is dropped into a crevasse, how fast will it be going just before it hits bottom 30 sec later?

Theory and Examples

55. **The geometric mean of a and b** The *geometric mean* of two positive numbers a and b is the number $\sqrt{ab}$. Show that the value of c in the conclusion of the Mean Value Theorem for $f(x) = 1/x$ on an interval of positive numbers $[a, b]$ is $c = \sqrt{ab}$.
56. **The arithmetic mean of a and b** The *arithmetic mean* of two numbers a and b is the number $(a + b)/2$. Show that the value of c in the conclusion of the Mean Value Theorem for $f(x) = x^2$ on any interval $[a, b]$ is $c = (a + b)/2$.

T 57. Graph the function

$$f(x) = \sin x \sin(x + 2) - \sin^2(x + 1).$$

What does the graph do? Why does the function behave this way? Give reasons for your answers.

58. Rolle's Theorem

- Construct a polynomial $f(x)$ that has zeros at $x = -2, -1, 0, 1$, and 2 .
 - Graph f and its derivative f' together. How is what you see related to Rolle's Theorem?
 - Do $g(x) = \sin x$ and its derivative g' illustrate the same phenomenon as f and f' ?
59. **Unique solution** Assume that f is continuous on $[a, b]$ and differentiable on (a, b) . Also assume that $f(a)$ and $f(b)$ have opposite signs and that $f' \neq 0$ between a and b . Show that $f(x) = 0$ exactly once between a and b .

60. **Parallel tangents** Assume that f and g are differentiable on $[a, b]$ and that $f(a) = g(a)$ and $f(b) = g(b)$. Show that there is at least one point between a and b where the tangents to the graphs of f and g are parallel or the same line. Illustrate with a sketch.
61. Suppose that $f'(x) \leq 1$ for $1 \leq x \leq 4$. Show that $f(4) - f(1) \leq 3$.
62. Suppose that $0 < f'(x) < 1/2$ for all x -values. Show that $f(-1) < f(1) < 2 + f(-1)$.
63. Show that $|\cos x - 1| \leq |x|$ for all x -values. (Hint: Consider $f(t) = \cos t$ on $[0, x]$.)
64. Show that for any numbers a and b , the sine inequality $|\sin b - \sin a| \leq |b - a|$ is true.
65. If the graphs of two differentiable functions $f(x)$ and $g(x)$ start at the same point in the plane and the functions have the same rate of change at every point, do the graphs have to be identical? Give reasons for your answer.
66. If $|f(w) - f(x)| \leq |w - x|$ for all values w and x and f is a differentiable function, show that $-1 \leq f'(x) \leq 1$ for all x -values.
67. Assume that f is differentiable on $a \leq x \leq b$ and that $f(b) < f(a)$. Show that f' is negative at some point between a and b .
68. Let f be a function defined on an interval $[a, b]$. What conditions could you place on f to guarantee that

$$\min f' \leq \frac{f(b) - f(a)}{b - a} \leq \max f',$$

where $\min f'$ and $\max f'$ refer to the minimum and maximum values of f' on $[a, b]$? Give reasons for your answers.

- T 69. Use the inequalities in Exercise 68 to estimate $f(0.1)$ if $f'(x) = 1/(1 + x^4 \cos x)$ for $0 \leq x \leq 0.1$ and $f(0) = 1$.
- T 70. Use the inequalities in Exercise 68 to estimate $f(0.1)$ if $f'(x) = 1/(1 - x^4)$ for $0 \leq x \leq 0.1$ and $f(0) = 2$.
71. Let f be differentiable at every value of x and suppose that $f(1) = 1$, that $f' < 0$ on $(-\infty, 1)$, and that $f' > 0$ on $(1, \infty)$.
- Show that $f(x) \geq 1$ for all x .
 - Must $f'(1) = 0$? Explain.
72. Let $f(x) = px^2 + qx + r$ be a quadratic function defined on a closed interval $[a, b]$. Show that there is exactly one point c in (a, b) at which f satisfies the conclusion of the Mean Value Theorem.

4.3

Monotonic Functions and the First Derivative Test

In sketching the graph of a differentiable function it is useful to know where it increases (rises from left to right) and where it decreases (falls from left to right) over an interval. This section gives a test to determine where it increases and where it decreases. We also show how to test the critical points of a function to identify whether local extreme values are present.

Increasing Functions and Decreasing Functions

As another corollary to the Mean Value Theorem, we show that functions with positive derivatives are increasing functions and functions with negative derivatives are decreasing functions. A function that is increasing or decreasing on an interval is said to be **monotonic** on the interval.

COROLLARY 3 Suppose that f is continuous on $[a, b]$ and differentiable on (a, b) .

If $f'(x) > 0$ at each point $x \in (a, b)$, then f is increasing on $[a, b]$.

If $f'(x) < 0$ at each point $x \in (a, b)$, then f is decreasing on $[a, b]$.

Proof Let x_1 and x_2 be any two points in $[a, b]$ with $x_1 < x_2$. The Mean Value Theorem applied to f on $[x_1, x_2]$ says that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

for some c between x_1 and x_2 . The sign of the right-hand side of this equation is the same as the sign of $f'(c)$ because $x_2 - x_1$ is positive. Therefore, $f(x_2) > f(x_1)$ if f' is positive on (a, b) and $f(x_2) < f(x_1)$ if f' is negative on (a, b) . ■

Corollary 3 is valid for infinite as well as finite intervals. To find the intervals where a function f is increasing or decreasing, we first find all of the critical points of f . If $a < b$ are two critical points for f , and if the derivative f' is continuous but never zero on the interval (a, b) , then by the Intermediate Value Theorem applied to f' , the derivative must be everywhere positive on (a, b) , or everywhere negative there. One way we can determine the sign of f' on (a, b) is simply by evaluating the derivative at a single point c in (a, b) . If $f'(c) > 0$, then $f'(x) > 0$ for all x in (a, b) so f is increasing on $[a, b]$ by Corollary 3; if $f'(c) < 0$, then f is decreasing on $[a, b]$. The next example illustrates how we use this procedure.

EXAMPLE 1 Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the intervals on which f is increasing and on which f is decreasing.

Solution The function f is everywhere continuous and differentiable. The first derivative

$$\begin{aligned} f'(x) &= 3x^2 - 12 = 3(x^2 - 4) \\ &= 3(x + 2)(x - 2) \end{aligned}$$

is zero at $x = -2$ and $x = 2$. These critical points subdivide the domain of f to create nonoverlapping open intervals $(-\infty, -2)$, $(-2, 2)$, and $(2, \infty)$ on which f' is either positive or negative. We determine the sign of f' by evaluating f' at a convenient point in each subinterval. The behavior of f is determined by then applying Corollary 3 to each subinterval. The results are summarized in the following table, and the graph of f is given in Figure 4.20.

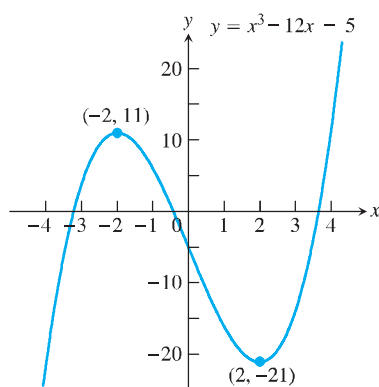


FIGURE 4.20 The function $f(x) = x^3 - 12x - 5$ is monotonic on three separate intervals (Example 1).

Interval	$-\infty < x < -2$	$-2 < x < 2$	$2 < x < \infty$
f' evaluated	$f'(-3) = 15$	$f'(0) = -12$	$f'(3) = 15$
Sign of f'	+	-	+
Behavior of f	increasing	decreasing	increasing

HISTORICAL BIOGRAPHY

Edmund Halley
(1656–1742)

We used “strict” less-than inequalities to specify the intervals in the summary table for Example 1. Corollary 3 says that we could use $\leq$ inequalities as well. That is, the function f in the example is increasing on $-\infty < x \leq -2$, decreasing on $-2 \leq x \leq 2$, and increasing on $2 \leq x < \infty$. We do not talk about whether a function is increasing or decreasing at a single point.

First Derivative Test for Local Extrema

In Figure 4.21, at the points where f has a minimum value, $f' < 0$ immediately to the left and $f' > 0$ immediately to the right. (If the point is an endpoint, there is only one side to consider.) Thus, the function is decreasing on the left of the minimum value and it is increasing on its right. Similarly, at the points where f has a maximum value, $f' > 0$ immediately to the left and $f' < 0$ immediately to the right. Thus, the function is increasing on the left of the maximum value and decreasing on its right. In summary, at a local extreme point, the sign of $f'(x)$ changes.

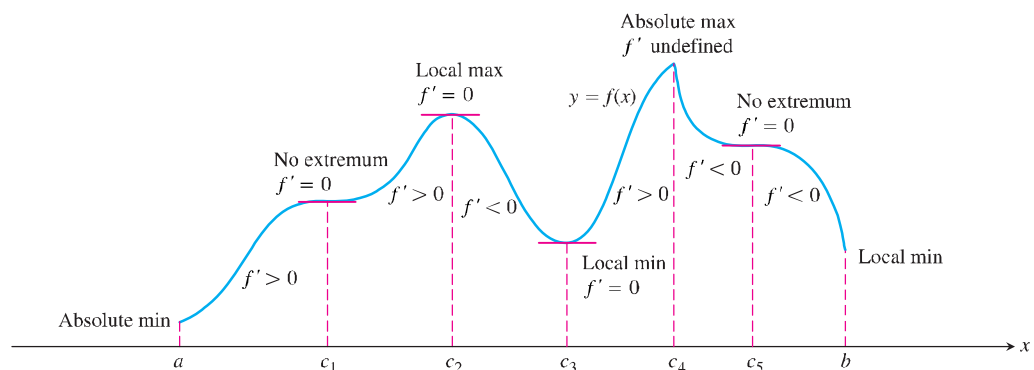


FIGURE 4.21 The critical points of a function locate where it is increasing and where it is decreasing. The first derivative changes sign at a critical point where a local extremum occurs.

These observations lead to a test for the presence and nature of local extreme values of differentiable functions.

First Derivative Test for Local Extrema

Suppose that c is a critical point of a continuous function f , and that f is differentiable at every point in some interval containing c except possibly at c itself. Moving across this interval from left to right,

1. if f' changes from negative to positive at c , then f has a local minimum at c ;
2. if f' changes from positive to negative at c , then f has a local maximum at c ;
3. if f' does not change sign at c (that is, f' is positive on both sides of c or negative on both sides), then f has no local extremum at c .

The test for local extrema at endpoints is similar, but there is only one side to consider.

Proof of the First Derivative Test Part (1). Since the sign of f' changes from negative to positive at c , there are numbers a and b such that $a < c < b$, $f' < 0$ on (a, c) , and $f' > 0$ on (c, b) . If $x \in (a, c)$, then $f(c) < f(x)$ because $f' < 0$ implies that f is decreasing on $[a, c]$. If $x \in (c, b)$, then $f(c) < f(x)$ because $f' > 0$ implies that f is increasing on $[c, b]$. Therefore, $f(x) \geq f(c)$ for every $x \in (a, b)$. By definition, f has a local minimum at c .

Parts (2) and (3) are proved similarly. ■

EXAMPLE 2 Find the critical points of

$$f(x) = x^{1/3}(x - 4) = x^{4/3} - 4x^{1/3}.$$

Identify the intervals on which f is increasing and decreasing. Find the function's local and absolute extreme values.

Solution The function f is continuous at all x since it is the product of two continuous functions, $x^{1/3}$ and $(x - 4)$. The first derivative

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^{4/3} - 4x^{1/3}) = \frac{4}{3}x^{1/3} - \frac{4}{3}x^{-2/3} \\ &= \frac{4}{3}x^{-2/3}(x - 1) = \frac{4(x - 1)}{3x^{2/3}} \end{aligned}$$

is zero at $x = 1$ and undefined at $x = 0$. There are no endpoints in the domain, so the critical points $x = 0$ and $x = 1$ are the only places where f might have an extreme value.

The critical points partition the x -axis into intervals on which f' is either positive or negative. The sign pattern of f' reveals the behavior of f between and at the critical points, as summarized in the following table.

Interval	$x < 0$	$0 < x < 1$	$x > 1$
Sign of f'	—	—	+
Behavior of f	decreasing	decreasing	increasing

Corollary 3 to the Mean Value Theorem tells us that f decreases on $(-\infty, 0]$, decreases on $[0, 1]$, and increases on $[1, \infty)$. The First Derivative Test for Local Extrema tells us that f does not have an extreme value at $x = 0$ (f' does not change sign) and that f has a local minimum at $x = 1$ (f' changes from negative to positive).

The value of the local minimum is $f(1) = 1^{1/3}(1 - 4) = -3$. This is also an absolute minimum since f is decreasing on $(-\infty, 1]$ and increasing on $[1, \infty)$. Figure 4.22 shows this value in relation to the function's graph.

Note that $\lim_{x \rightarrow 0} f'(x) = -\infty$, so the graph of f has a vertical tangent at the origin.

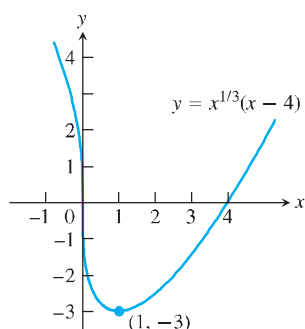


FIGURE 4.22 The function $f(x) = x^{1/3}(x - 4)$ decreases when $x < 1$ and increases when $x > 1$ (Example 2).

Exercises 4.3

Analyzing Functions from Derivatives

Answer the following questions about the functions whose derivatives are given in Exercises 1–14:

- What are the critical points of f ?
- On what intervals is f increasing or decreasing?
- At what points, if any, does f assume local maximum and minimum values?

- $f'(x) = x(x - 1)$
- $f'(x) = (x - 1)(x + 2)$
- $f'(x) = (x - 1)^2(x + 2)$
- $f'(x) = (x - 1)^2(x + 2)^2$
- $f'(x) = (x - 1)(x + 2)(x - 3)$
- $f'(x) = (x - 7)(x + 1)(x + 5)$
- $f'(x) = \frac{x^2(x - 1)}{x + 2}, \quad x \neq -2$

$$8. f'(x) = \frac{(x - 2)(x + 4)}{(x + 1)(x - 3)}, \quad x \neq -1, 3$$

$$9. f'(x) = 1 - \frac{4}{x^2}, \quad x \neq 0 \quad 10. f'(x) = 3 - \frac{6}{\sqrt{x}}, \quad x \neq 0$$

$$11. f'(x) = x^{-1/3}(x + 2) \quad 12. f'(x) = x^{-1/2}(x - 3)$$

$$13. f'(x) = (\sin x - 1)(2 \cos x + 1), \quad 0 \leq x \leq 2\pi$$

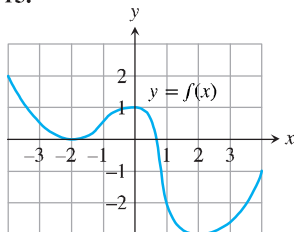
$$14. f'(x) = (\sin x + \cos x)(\sin x - \cos x), \quad 0 \leq x \leq 2\pi$$

Identifying Extrema

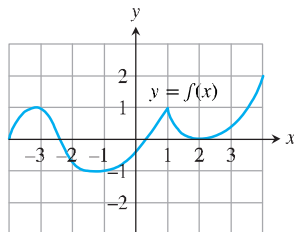
In Exercises 15–40:

- Find the open intervals on which the function is increasing and decreasing.
- Identify the function's local and absolute extreme values, if any, saying where they occur.

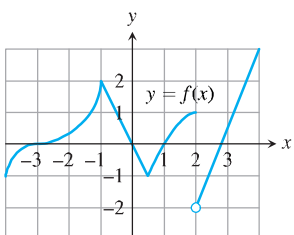
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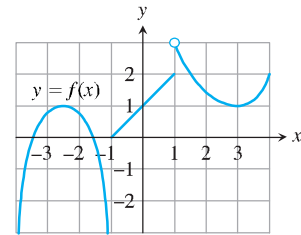
16.



17.



18.



19. $g(t) = -t^2 - 3t + 3$

20. $g(t) = -3t^2 + 9t + 5$

21. $h(x) = -x^3 + 2x^2$

22. $h(x) = 2x^3 - 18x$

23. $f(\theta) = 3\theta^2 - 4\theta^3$

24. $f(\theta) = 6\theta - \theta^3$

25. $f(r) = 3r^3 + 16r$

26. $h(r) = (r + 7)^3$

27. $f(x) = x^4 - 8x^2 + 16$

28. $g(x) = x^4 - 4x^3 + 4x^2$

29. $H(t) = \frac{3}{2}t^4 - t^6$

30. $K(t) = 15t^3 - t^5$

31. $f(x) = x - 6\sqrt{x-1}$

32. $g(x) = 4\sqrt{x} - x^2 + 3$

33. $g(x) = x\sqrt{8-x^2}$

34. $g(x) = x^2\sqrt{5-x}$

35. $f(x) = \frac{x^2-3}{x-2}, \quad x \neq 2$

36. $f(x) = \frac{x^3}{3x^2+1}$

37. $f(x) = x^{1/3}(x+8)$

38. $g(x) = x^{2/3}(x+5)$

39. $h(x) = x^{1/3}(x^2-4)$

40. $k(x) = x^{2/3}(x^2-4)$

In Exercises 41–52:

- a. Identify the function's local extreme values in the given domain, and say where they occur.

- b. Which of the extreme values, if any, are absolute?

T c. Support your findings with a graphing calculator or computer grapher.

41. $f(x) = 2x - x^2, \quad -\infty < x \leq 2$

42. $f(x) = (x+1)^2, \quad -\infty < x \leq 0$

43. $g(x) = x^2 - 4x + 4, \quad 1 \leq x < \infty$

44. $g(x) = -x^2 - 6x - 9, \quad -4 \leq x < \infty$

45. $f(t) = 12t - t^3, \quad -3 \leq t < \infty$

46. $f(t) = t^3 - 3t^2, \quad -\infty < t \leq 3$

47. $h(x) = \frac{x^3}{3} - 2x^2 + 4x, \quad 0 \leq x < \infty$

48. $k(x) = x^3 + 3x^2 + 3x + 1, \quad -\infty < x \leq 0$

49. $f(x) = \sqrt{25-x^2}, \quad -5 \leq x \leq 5$

50. $f(x) = \sqrt{x^2-2x-3}, \quad 3 \leq x < \infty$

51. $g(x) = \frac{x-2}{x^2-1}, \quad 0 \leq x < 1$

52. $g(x) = \frac{x^2}{4-x^2}, \quad -2 < x \leq 1$

In Exercises 53–60:

- a. Find the local extrema of each function on the given interval, and say where they occur.

T b. Graph the function and its derivative together. Comment on the behavior of f in relation to the signs and values of f' .

53. $f(x) = \sin 2x, \quad 0 \leq x \leq \pi$

54. $f(x) = \sin x - \cos x, \quad 0 \leq x \leq 2\pi$

55. $f(x) = \sqrt{3} \cos x + \sin x, \quad 0 \leq x \leq 2\pi$

56. $f(x) = -2x + \tan x, \quad \frac{-\pi}{2} < x < \frac{\pi}{2}$

57. $f(x) = \frac{x}{2} - 2 \sin \frac{x}{2}, \quad 0 \leq x \leq 2\pi$

58. $f(x) = -2 \cos x - \cos^2 x, \quad -\pi \leq x \leq \pi$

59. $f(x) = \csc^2 x - 2 \cot x, \quad 0 < x < \pi$

60. $f(x) = \sec^2 x - 2 \tan x, \quad \frac{-\pi}{2} < x < \frac{\pi}{2}$

Theory and Examples

Show that the functions in Exercises 61 and 62 have local extreme values at the given values of θ , and say which kind of local extreme the function has.

61. $h(\theta) = 3 \cos \frac{\theta}{2}, \quad 0 \leq \theta \leq 2\pi, \quad \text{at } \theta = 0 \text{ and } \theta = 2\pi$

62. $h(\theta) = 5 \sin \frac{\theta}{2}, \quad 0 \leq \theta \leq \pi, \quad \text{at } \theta = 0 \text{ and } \theta = \pi$

63. Sketch the graph of a differentiable function $y = f(x)$ through the point $(1, 1)$ if $f'(1) = 0$ and

- a. $f'(x) > 0$ for $x < 1$ and $f'(x) < 0$ for $x > 1$;
 b. $f'(x) < 0$ for $x < 1$ and $f'(x) > 0$ for $x > 1$;
 c. $f'(x) > 0$ for $x \neq 1$;
 d. $f'(x) < 0$ for $x \neq 1$.

64. Sketch the graph of a differentiable function $y = f(x)$ that has

- a. a local minimum at $(1, 1)$ and a local maximum at $(3, 3)$;
 b. a local maximum at $(1, 1)$ and a local minimum at $(3, 3)$;
 c. local maxima at $(1, 1)$ and $(3, 3)$;
 d. local minima at $(1, 1)$ and $(3, 3)$.

65. Sketch the graph of a continuous function $y = g(x)$ such that

- a. $g(2) = 2, 0 < g' < 1$ for $x < 2, g'(x) \rightarrow 1^-$ as $x \rightarrow 2^-$,
 $-1 < g' < 0$ for $x > 2$, and $g'(x) \rightarrow -1^+$ as $x \rightarrow 2^+$;
 b. $g(2) = 2, g' < 0$ for $x < 2, g'(x) \rightarrow -\infty$ as $x \rightarrow 2^-$,
 $g' > 0$ for $x > 2$, and $g'(x) \rightarrow \infty$ as $x \rightarrow 2^+$.

66. Sketch the graph of a continuous function $y = h(x)$ such that

- a. $h(0) = 0, -2 \leq h(x) \leq 2$ for all $x, h'(x) \rightarrow \infty$ as $x \rightarrow 0^-$,
and $h'(x) \rightarrow \infty$ as $x \rightarrow 0^+$;
 b. $h(0) = 0, -2 \leq h(x) \leq 0$ for all $x, h'(x) \rightarrow \infty$ as $x \rightarrow 0^-$,
and $h'(x) \rightarrow -\infty$ as $x \rightarrow 0^+$.

67. Discuss the extreme-value behavior of the function $f(x) = x \sin(1/x), x \neq 0$. How many critical points does this function have? Where are they located on the x -axis? Does f have an

absolute minimum? An absolute maximum? (See Exercise 49 in Section 2.3.)

68. Find the intervals on which the function $f(x) = ax^2 + bx + c$, $a \neq 0$, is increasing and decreasing. Describe the reasoning behind your answer.

69. Determine the values of constants a and b so that $f(x) = ax^2 + bx$ has an absolute maximum at the point $(1, 2)$.

70. Determine the values of constants a , b , c , and d so that $f(x) = ax^3 + bx^2 + cx + d$ has a local maximum at the point $(0, 0)$ and a local minimum at the point $(1, -1)$.

4.4

Concavity and Curve Sketching

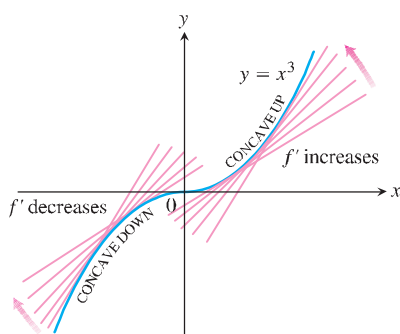


FIGURE 4.23 The graph of $f(x) = x^3$ is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$ (Example 1a).

We have seen how the first derivative tells us where a function is increasing, where it is decreasing, and whether a local maximum or local minimum occurs at a critical point. In this section we see that the second derivative gives us information about how the graph of a differentiable function bends or turns. With this knowledge about the first and second derivatives, coupled with our previous understanding of asymptotic behavior and symmetry studied in Sections 2.6 and 1.1, we can now draw an accurate graph of a function. By organizing all of these ideas into a coherent procedure, we give a method for sketching graphs and revealing visually the key features of functions. Identifying and knowing the locations of these features is of major importance in mathematics and its applications to science and engineering, especially in the graphical analysis and interpretation of data.

Concavity

As you can see in Figure 4.23, the curve $y = x^3$ rises as x increases, but the portions defined on the intervals $(-\infty, 0)$ and $(0, \infty)$ turn in different ways. As we approach the origin from the left along the curve, the curve turns to our right and falls below its tangents. The slopes of the tangents are decreasing on the interval $(-\infty, 0)$. As we move away from the origin along the curve to the right, the curve turns to our left and rises above its tangents. The slopes of the tangents are increasing on the interval $(0, \infty)$. This turning or bending behavior defines the *concavity* of the curve.

DEFINITION The graph of a differentiable function $y = f(x)$ is

- (a) **concave up** on an open interval I if f' is increasing on I ;
- (b) **concave down** on an open interval I if f' is decreasing on I .

If $y = f(x)$ has a second derivative, we can apply Corollary 3 of the Mean Value Theorem to the first derivative function. We conclude that f' increases if $f'' > 0$ on I , and decreases if $f'' < 0$.

The Second Derivative Test for Concavity

Let $y = f(x)$ be twice-differentiable on an interval I .

1. If $f'' > 0$ on I , the graph of f over I is concave up.
2. If $f'' < 0$ on I , the graph of f over I is concave down.

If $y = f(x)$ is twice-differentiable, we will use the notations f'' and y'' interchangeably when denoting the second derivative.